Many known numbers, one reading: a census of the constants touched by a two-mirror geometry, with a registered prediction for the dark-to-baryon ratio

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Abstract

Standard physics leaves many numbers free. This record lists the known numbers touched by the readings of the author's framework, Mathematical Materiality (MM), and asks which of them MM reads in a form that no other work we found gives. The work goes from the data to MM. A reading chosen after looking at the data is admitted if its choices are declared; only a fixed reading applied to data not yet seen is called a prediction. (1) One rule, two distant numbers. A single geometric factor, √3 per spatial direction between two mirror images, reads the cosmic dark-to-baryon ratio Ω_c/Ω_b = 3√3 = 5.196 and the galactic acceleration scale $a_0 = cH_0/6$. The two most recent CMB combinations, which share data, read 3√3 within 2σ: P-ACT-LB2 gives 5.199 and CMB-SPA + DESI DR2 gives 5.254 ± 0.037, with the correlations of the published chains. Planck 2018 alone is at 2.1σ. SPARC reads $cH_0/6$ within its systematic uncertainty. The reading was chosen knowing these values and its choices are declared: it is a coherent reading, not a derivation.

Version 2.54 of the Mathematical Materiality (MM) series, October 2026. Continues MM 2.44 (consistency checks of Ω_c/Ω_b = 3√3 against Planck and DESI DR2, DOI 10.5281/zenodo.23329080) and references MM 2.46 (the galaxy scale $cH_0/6$ in clusters, DOI 10.5281/zenodo.23340834).

In the working records this material is in the folder «MM_sviluppo_NUMERO» («the distinctive number»); working files are in Italian. The prediction was registered under the working names «previsione v1 … v5».

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1. What this deposit is, and what it is not

The question. Standard physics leaves many numbers free: they are measured, not explained. MM is a framework in which matter is a reading of correlated copies, and several of its readings use mirror images. This deposit asks three things, in this order:

1. which known numbers do MM readings touch, and with which measurements are they compatible;
2. which of them does MM read in a form that no other work we found gives;
3. how can that form be tested on data that do not exist yet.

The work goes **from the data to MM**: a reading chosen after looking at the data is admitted if its choices are declared (level 1, «coherent reading»); only a fixed reading applied to data not yet seen is called a prediction (level 2).

The results.

- **One rule, two distant numbers.** A single geometric factor, √3 per spatial direction between two mirror images, reads the cosmic dark-to-baryon ratio, **Ω_c/Ω_b = 3√3 = 5.196**, and the galactic acceleration scale, **$a_0 = cH_0/6$**.
- The two most recent CMB combinations read 3√3 within 2σ: P-ACT-LB2 gives 5.199 ± 0.040, and CMB-SPA + DESI DR2 gives 5.254 ± 0.037 (1.6σ), with the correlations of the published chains. The two combinations share data. Planck 2018 alone is at 2.1σ.
- SPARC reads $cH_0/6$ within its systematic uncertainty.
- The reading was chosen knowing these values: it is a coherent reading, not a derivation.
- **The census.** Section 2 is a selection from a working catalogue that is still partial (the extraction of the deposits 2.0-2.32 is not complete).

- It has 18 entries that are compatible with the measurements, or not excluded, under the conditions stated in each row, and 7 entries with a future check. The entries are not 25 independent numbers: one row rewrites another, one groups several structures, and some reappear among the future checks.
- The unfavourable outcomes are reported too.
- Where standard physics or another author already gives the same number or the same relation, the table says so: there MM is compatible, not distinctive.
- **The distinctive form.** Among the precedents we identified, the only work that gives both relations is Verlinde's emergent gravity (2016).
- **His two relations:** a_M = a_0/6 with $a_0 = cH_0$, and Ω_D² = (4/3)Ω_B (his eq. 7.51). Verlinde presents the second as a conjectural extension («a final fun comment»). Here it is taken as exact, with Ω_D identified with Ω_c.
- **What it gives:** ω_c/ω_b = 2h/√(3ω_b). This equals 3√3 only when Ω_b = 4/81, i.e. for h = h* = (9/2)√ω_b, about 0.675 for the ω_b values used (0.6730-0.6762).
- **The difference:** in MM the ratio is 3√3 **for any h**. Today h is close to h*, and the published data do not separate the two forms (section 4).
- **A registered prediction.** Two data sets may decide between the two forms, or leave the comparison open: the first DESI BAO release beyond DR2 and the first Simons Observatory parameters combined with DESI. The rules for them are fixed in advance and applied by a program (section 5).
- Only articles posted after 16 October 2026, 02:22:54 UTC are admitted.
- A query of the arXiv API at 01:49 UTC that day found no such article among those indexed.

What it is not.

- It is not a derivation. Many entries of the census coincide with standard physics (γ = β = 1, the Tsirelson bound) or with relations proposed by other authors (Koide 1981, Brannen 2006, Rivero 2026, Verlinde 2016).
- The strong coupling is not read. α_s(M_Z) enters as data, like the masses; the ratio R is built from measured α_s and α_2 by the Standard Model running; and any reading that returns α_s through R is discarded by construction (documenti/forte/). Of the readings that output α_s, the 28 registered counting mechanisms and 16π fell. Two non-circular relations, both chosen after looking at the data, remain:
- $\alpha_3 = (\alpha_2 + 2\alpha_Y)/3$ at the Planck scale (residual dependence on α_s through the running: 1.9 %). Its fixed point is 0.11625 with the PDG 2025 inputs and 0.11666 with the PDG 2026 inputs: z = −1.08 against the world average, and z = −2.02 with the ALPHA 2026 lattice value as input (tension, not exclusion). It is the relation submitted to a registered prospective test on the next PDG edition and the next FLAG review (documenti/forte/test_G2/), where a compatibility counts as precise only if σ_tot ≤ 0.070;
- Y/W = (F/W)^(3/2), in tension with the ALPHA 2026 value (3.3σ from the input errors alone); not tested further.
- It is not a test of complete cosmologies. The parameters are those inferred within ΛCDM by the collaborations; the test compares two relations among them.
- A favourable outcome would be an A-confirmation on a parameter already used to choose the reading. It would support the MM reading of the ratio in the domain examined. It would not confirm MM as a whole, prove that the ratio is independent of h for every h, or prove that no other theory gives the same form.

2. The census

From the Italian working catalogue (documenti/censimento/CATALOGO_NUMERI_v3.md, registered 16.10.2026 00:34:02 UTC), which collects the deposits 2.34-2.52, the book NUMERI and, partially, the deposits 2.0-2.32. «Also given by» states where MM coincides with standard physics or with another author. $R = (1/\alpha_3)/(1/\alpha_2)$ is the ratio of the inverse strong and weak couplings at the Planck scale, an input of the readings that use it: 1.0768 ± 0.0014 with the two-loop running of MM 2.38, 1.0761 ± 0.0013 with the three-loop running of codice/corsa3.py (PDG 2025 inputs). The Higgs row uses the first value, the CKM and neutron rows the second; with the other value each entry moves by less than its error. Every number of this section that can be recomputed from public inputs is recomputed by codice/ricalcolo_censimento_v2.py (section 4, Checks before publication).

2.1 Compatible with the measurements, or not excluded, under the stated conditions

number read by MM	measured	also given by	conditions
Ω_c/Ω_b = 3√3 = 5.196	P-ACT-LB2 5.199 ± 0.040; CMB-SPA + DESI DR2 5.254 ± 0.037 (1.6σ); Planck 2018 alone 5.364 ± 0.079 (2.1σ); errors with the chain correlations	Verlinde 2016, if Ω_b = 4/81	two-mirror reading; ΛCDM parameters
$a_0 = cH_0/6 = (1.09\text{-}1.18)\cdot10^{-10}$ m s^{-2}	SPARC $(1.20 \pm 0.02 \pm 0.24)\cdot10^{-10}$	Verlinde 2016 (a_M = cH_0/6); noted as an empirical	same rule; within the systematic

		value, $a_0 \approx cH_0/6$, by Sanders (arXiv:1811.05260, 1912.00716)	uncertainty
cluster baryon fraction $1/(1 + 3\sqrt{3}) = 0.161$	X-COP 0.141-0.165 at R500	—	the same ratio, rewritten; a comparison cluster by cluster needs a model of radius, depletion, stars and gas
shape of $a_0(z)$: $a_0(z)/a_0(z_1) = E(z)/E(z_1)$, $E = H/H_0$	MUSE-DARK III, four bins, within 1σ	Milgrom 2015	registered prospective check
Koide $Q = 2/3$ (charged leptons)	0.6666645 ± 0.0000051 (0.4σ)	Koide 1981	MM gives a reading of the relation, not the number
down quarks through the mirror, Q(1/m)	0.6647 (PDG masses) / 0.6674 (at M_Z)	Rivero 2026	within uncertainties
$\Sigma m_\nu = 0.0593$ eV, normal ordering	ΛCDM bounds: < 0.064 (DESI DR2 + CMB), < 0.071-0.077 (PDG 2025) allow it; < 0.048 (with SPT-3G) and < 0.053 (with Lyman-α) exclude it, together with the whole normal ordering (minimum 0.0589); w0waCDM < 0.186	the same relation was proposed by Brannen 2006 (0.0600 eV with the inputs of the time)	allowed by the bounds that allow the normal ordering; a value near 0.06 eV would not single out this relation
$M_W \approx 80.379$ GeV from the pair (4, −2)	80.3692 ± 0.0133 GeV (0.7σ)	—	the joint hypothesis (4, −2) is not validated
Higgs self-coupling $\lambda(M_P) = -(R - 1)/(4\pi) = -0.0061$	−0.0100 ± 0.0031 (1.3σ)	$\lambda \approx 0$: Froggatt–Nielsen 1995; Shaposhnikov–Wetterich 2010	R is an input built from measured α_s and α_2; the measured $\lambda(M_P)$ also uses α_s in its running, so the two sides have correlated errors and the comparison needs their covariance
first-row CKM deficit $\delta^2/2 = 0.067$ %	0.17 ± 0.07 % (1.5σ)	no prior work on the value found	R is an input built from measured α_s and α_2; the target does not contain α_s; the measured deficit depends on V_{us}
$\alpha_s(M_Z) = 0.1180$ with $c = 1/\sqrt{3}$ (levels)	0.1180 ± 0.0009	—	reading built on the data
$\gamma = \beta = 1$	Cassini: $\gamma - 1 = (2.1 \pm 2.3)\cdot 10^{-5}$	general relativity	only if R is a pure ratio of units
speed of gravitational waves = c	GW170817	general relativity	same as GR
lensing slip $\eta = 1$	KiDS 0.916 [0.786; 1.056]	general relativity	same as GR
Tsirelson bound $2\sqrt{2}$	2.82759 ± 0.00051	quantum mechanics	same as QM (MM 2.48)
three-path interference $I_3 = 0$	$< 10^{-2}$ (Sinha 2010)	quantum mechanics (Sorkin)	same as QM
maximum compactness C = 2^(1/3) − 2^(−1/3) = 0.46622	NICER: compatible, not distinguishable	limit 4/9 for a perfect fluid: Buchdahl 1959; area variable gives 1/2: Jacobson 2007	ideal uniform-density model with the isotropic variable (MM 2.34); mathematical, causal and stable limits differ
spin ½, two photon polarisations, eight gluons, three colours	½, 2, 8, 3	D'Ariano–Perinotti; Wilson; Pati–Salam	structures (MM 2.40, 2.52)

2.2 Entries with a future check

reading	today	possible check, and when
$3\sqrt{3}$ for any h (against Verlinde)	not separated (section 4)	DESI BAO beyond DR2 with the CMB; Simons Observatory

		with DESI (section 5)
shape of $a_0(z)$	holds at 1σ	new samples at z = 1-2 (prediction registered on 12.10.2026 in the working records)
wide binaries are Newtonian	Gaia DR3: R ≈ 1.25 in our analysis, moderate tension	Gaia DR4
$\alpha_s(M_Z)$ from $\alpha_3 = (\alpha_2 + 2\alpha_Y)/3$ at the Planck scale (form chosen after the data): 0.11625 (PDG 2025 inputs), 0.11666 (PDG 2026)	z = −1.08 against the world average; z = −2.02 with the ALPHA 2026 lattice value as input (tension, not exclusion); not circular (documenti/forte/)	registered prospective test on the next PDG edition and the next FLAG review (documenti/forte/test_G2/, version 3); a compatibility is precise only if $\sigma_{tot} \leq 0.070$; with today's other inputs a lattice value of 0.1188 would give tension, not exclusion
neutron beam/bottle difference δ/3 = 1.22 %	1.17 ± 0.23 %, but no rule chooses between the two readings that give −0.2σ and −5.7σ	BL3, UCNProBe
first-row CKM deficit	1.5σ	PIONEER (at 0.05 % alone it does not discriminate the predicted deficit)
Σm_ν and the ordering	allowed by the bounds that allow the normal ordering	Simons Observatory + DESI, JUNO
maximum neutron-star mass 5 % above GR	degenerate with the equation of state	NICER, eXTP

2.3 Unfavourable outcomes

Readings that fell: the strong coupling from 28 registered counting mechanisms (0 hits in 28); a «lit fraction» of 0.1671 against 0.1571 (5.1σ from Planck 2018 with the chain correlation, 7.6σ from CMB-SPA + DESI DR2); Ω_Λ from rhythm × momentum (0.483 or 0.820 against 0.698); the «side-by-side cubes» law, which gives 2.43 at z = 1100; «D = 3 selected» by the two-wall invariant; NGC 2419 (ν = 2.75-3.06 predicted against $\lesssim$ 1.1 required).

Withdrawn: $w \geq -1$ as an MM number; the +43.9 nat on SPARC; with the 30 pc cell, which is a postulate (erratum to MM 2.23), the globular clusters and the Coma cluster (Saturn stays compatible with the cell).

Open problems or unfavourable comparisons still to be qualified:

- **Clusters:** the acceleration scale $g\ddagger = 2.02 \cdot 10^{-9}$ m s^{-2} is 16.7 ± 1.2 times the galaxy value, and the two-mirror rule does not read it.
- **Diffuse dwarfs:** with the external field they come out short by a factor 1.7-3.8 in σ.
- **Palomar 14:** 0.79-1.11 km/s predicted against 0.38 ± 0.12 km/s measured.
- **Dwarf spheroidals and ultra-faint dwarfs:** not read without a law for each system.

The outcomes taken from the extracts of the deposits 2.0-2.32 are to be reconstructed before any further use. Each unfavourable outcome concerns one reading, not the framework.

Coincidences noticed and not used: $2\pi - 1 \approx 5.28$; $R - 1 \approx 1/13$; $cH_0/(2e) \approx 1.20 \cdot 10^{-10}$ m s^{-2}; the top mass with 27π; $\Omega_b = 4/81$, which makes $3\sqrt{3}$ equal to Verlinde's value.

3. The two-mirror reading (level 1)

The reading has five declared choices:

- **S1.** A factor $\ell = \sqrt{3}$ per spatial direction between the two images.
- **S2.** Our matter keeps its mass: a quantity containing h powers of length transforms as ℓ^h.
- **S3.** The content of the other image fills the volume: integrals scale as ℓ^3, so $\Omega_c/\Omega_b = \ell^3 = 3\sqrt{3}$.
- **S4.** The base acceleration is $cH_0/2$, from the balance $R_s/r = r/R_H$.
- **S5.** Read across two directions, it gives $a_0 = (cH_0/2) \cdot \ell^{-2} = cH_0/6$.

The choices were made knowing the data (level 1). With them the reading gives:

- two numerical relations, $3\sqrt{3}$ (P-ACT, CMB-SPA + DESI) and $cH_0/6$ (SPARC);
- the cluster baryon fraction, which is a rewriting of the first relation: a comparison with X-COP needs a cluster model;
- the shape of $a_0(z)$.

Without additions it does not read the MUSE-DARK III normalisation (16-27 % above), groups, clusters, dwarfs, globular clusters or single galaxies. Two AI reviewers (Codex/GPT-6 and Gemini) checked this reading at level 1 and gave «coherent with corrections»; the corrections were applied.

4. Test on the published data: MM against Verlinde

Registration. The procedure was registered before the calculation (documenti/test_oggi/PROCEDURA_LIVELLO2_K1.md, 01:10:49 UTC). Data, deviation and program were registered at 01:15:45 UTC (DATI_E_DEVIAZIONE.md, codice/test_mm_verlinde.py).

Method.

- **Correlations:** the published tables give none between ω_b, ω_c and h, so a grid of 32 correlation matrices was declared in advance. The published chains, downloaded afterwards, give the actual correlations (Checks, below).
- **Statistic:** χ^2 for each relation with linear error propagation, and $\Delta\chi^2 = \chi^2_\text{Verlinde} - \chi^2_\text{MM}$.
- **Zones,** fixed in advance: $|\Delta\chi^2| \leq 1$ no preference, 1-4 hint, > 4 preference.

combination	ω_c/ω_b measured	MM	Verlinde	Δχ²	zone
Planck 2018 (CMB only)	5.3643	5.1962	5.2035	−3.93 … −0.10	below or above the hint threshold depending on the correlations (always towards Verlinde)
P-ACT-LB2	5.1993	5.1962	5.2561	+1.02 … +2.35	hint towards MM
CMB-SPA + DESI DR2	5.2536	5.1962	5.2449	−2.80 … −1.85	hint towards Verlinde

Outcome. With the registered numbers and grid, no combination reaches a preference. The two most recent combinations share Planck, ACT DR6 and DESI DR2, and give hints in opposite directions. Today the test does not separate the two forms (grade B*: data already seen).

Checks (risultati/):

- **Monte Carlo propagation:** agrees with the linear one within 0.01.
- **More precise H_0 for P-ACT-LB2** (68.43 ± 0.27 from ACT DR6, Table 5): same zone.
- **Wider correlation grids:**
- With 6115 matrices, P-ACT-LB2 and CMB-SPA + DESI keep their hints, and Planck stays undecided.
- With the 7349 matrices of the prediction's grid, P-ACT-LB2 keeps its hint towards MM. CMB-SPA + DESI gives at least a hint towards Verlinde, with some matrices in the preference zone and some outside 2σ for MM.
- No grid gives a uniform preference.
- **Correlations from the published chains** (documenti/test_oggi/catene/; plan registered before the download, program before the calculation). Chains: Planck 2018 plikHM TT,TE,EE + lowE + lensing; CMB-SPA + DESI DR2 (the corrected SPT-3G release); for P-ACT-LB2, which has no public chain, the P-ACT-LB chain with DESI DR1 BAO, used for its correlations only. The chain means and errors agree with the tables (within 1σ and 20 %).

combination	ρ(ω_b, ω_c), ρ(ω_b, h), ρ(ω_c, h)	Δχ², registered grid	Δχ², chain correlations	zone
Planck 2018	−0.53, +0.71, −0.96 (outside the grid, mostly in ρ(ω_b, h))	−3.93 … −0.10	−2.15	hint towards Verlinde
P-ACT-LB2 (DR1 chain)	−0.26, +0.51, −0.93	+1.02 … +2.35	+1.09	hint towards MM

- Each combination now falls in a single zone, and each $\Delta\chi^2$ lies inside the range of the registered test (for Planck too, although its correlations are outside the grid). None reaches a preference, so the outcome is unchanged: today the test does not separate the two forms.
- Other burn-in fractions give the same zones.
- The chain archives (1.49 GB in all; sizes and SHA-256 in SOURCES.md) are not included. Their weighted moments are in dati/covarianze_catene.json, and codice/ricalcolo_catene.py recomputes the last step from them.

Checks before publication (16.10.2026; documenti/verifiche_publish/, each program registered before it ran):

- **The census recomputed:** codice/ricalcolo_censimento_v2.py recomputes every number of section 2 that can be redone from public inputs (PDG 2025, CODATA 2022, Buttazzo et al. eq. 61, the running of corsa3.py, the Planck chain). Its log marks five lines DIFFERENT. They belong to two corrections already in this text: the two values of R (three lines: R itself, and the CKM and neutron entries computed with it) and the error on f_b with the chain correlation, which gives 5.1σ instead of 6σ for the «lit fraction» (two lines). The check also gave M_W at 0.74σ instead of 0.8σ (corrected in the table) and an error of 0.0023 on the measured Y/W where the working catalogue has 0.0021. A third correction, the full list of the Σm_ν bounds, comes from the bounds themselves. All other recomputed numbers agree with the text at the digits written.
- **Prior work:** INSPIRE-HEP full text and arXiv abstracts. No earlier statement of $\Omega_c/\Omega_b = 3\sqrt{3}$ was found; the search for the symbol √ is unreliable on INSPIRE, and ADS and Google Scholar were not searched. For $a_0 = cH_0/6$, Sanders' empirical remark was found and added to the table.
- **The running:** codice/prova_corsa3_buttazzo_v2.py compares codice/corsa3.py with eq. 61 of Buttazzo et al. (2013): couplings at M_Pl within 0.00004 (λ within 0.00015), derivatives within 20 %, 10 checks out of 10. The effect on the G2 statistic is 0.009, below the rounding of the source.
- **Related records:** the two DOIs cited at the top resolve to MM 2.44 and MM 2.46, as described.
- **External reading:** Codex/GPT-6 read the corrections of this afternoon once: «ready with corrections», five points, all applied (documenti/verifiche_publish/VERIFICA_ASTRA/).

5. The registered prediction

The prediction went through five versions (documenti/previsione/, all registered). Version 5 (02:22:54 UTC) is the one in force; it is the text of version 4 plus the corrections of the fourth verification round. The rules are applied by codice/decidi_previsione_v5.py. If text and program disagree, the program prevails.

Targets. Only articles posted on arXiv after the registration of version 5 are admitted.

- **DESI block:** the first DESI BAO release beyond DR2, i.e. the collaboration paper that combines all BAO tracers of that release with the CMB.
- **Simons Observatory block:** the first SO paper with ΛCDM parameters that combine SO data with DESI BAO.
- **Freezing:** each block is frozen at T_c, the date of the first arXiv version plus 90 days, 00:00 UTC.
- **What enters:** the materials public at T_c, i.e. that version, the authors' errata issued before T_c, and the chains available at T_c.
- **Later errata:** an erratum issued after T_c leads to a registered update next to the first result.

Data.

- **Admissible combinations:** ΛCDM, CMB plus the data that define the target, nothing else.
- **Principal combination:** chosen by the program, in this order:
- among the admissible combinations the authors indicate as baseline, if any;
- otherwise among all admissible combinations;
- ties are broken by the smallest error on $\Omega_c h^2$, then on h, then by order in the paper.
- **Covariances:** taken from the published chains or, if there are none, from a fixed grid of 7349 correlation matrices.

Rules. All conditions hold on all matrices of the principal combination; outcomes can hold together.

- **Exclusion:** a relation falls if $\chi^2 > 9$.
- **Relative fall:** a relation also falls if all three conditions hold:
- its $\chi^2 > 4$;
- the other relation is preferred ($\Delta\chi^2$ beyond 4) and compatible ($\chi^2 \leq 4$);
- no single matrix of any other admissible combination gives $\Delta\chi^2$ beyond 1 in its favour.

- **A-confirmation of the MM reading** (on a parameter already used to choose it): all three conditions hold:
- MM is preferred ($\Delta\chi^2 > 4$);
- MM is compatible ($\chi^2 \leq 4$);
- no single matrix of any other admissible combination gives $\Delta\chi^2 < -1$.
- **«7.51 favoured»:** the symmetric outcome for Verlinde.
- **Otherwise** the test stays open. A block with no admissible combination is «not evaluable»; a malformed input file gives «invalid data», which is not a scientific outcome.

Combining the two blocks.

- **Falls stand.**
- **A confirmation of one block does not hold overall** if the other block makes that reading fall, gives the opposite outcome, or shows a hint, at least a hint, or a preference in the opposite direction in its principal combination.
- **A missing or non-evaluated block** keeps the overall result provisional.
- **Numbers made public before registration** lower either confirmation to grade B*, and it stays B* overall.

If the MM reading falls, what falls is the MM reading of the ratio, not MM.

When the test can separate. Let $k = 2/\sqrt{(3\omega_b)} \approx 7.70$ and $h^* = (9/2)\sqrt{\omega_b}$. Let σ_R be the uncertainty of the measured ratio $R = \omega_c/\omega_b$, and σ_D that of $R - P_V$, where P_V is Verlinde's value.

- A ratio measured on MM gives $\Delta\chi^2 = k^2(h - h^*)^2/\sigma_D^2$.
- A ratio measured on Verlinde gives $\Delta\chi^2 = -k^2(h - h^*)^2/\sigma_R^2$.
- An intermediate ratio can cancel $\Delta\chi^2$ or change its sign.

Today $h - h^*$ is +0.0010 (Planck), +0.0078 (P-ACT-LB2) and +0.0063 (CMB-SPA + DESI). With the CMB-SPA + DESI uncertainties on the 32-matrix grid of section 4, a separation over that whole grid needs $|h - h^*| > 0.0133$ (MM case) or > 0.0108 (Verlinde case); smaller uncertainties need less. This is a separation threshold, not a statistical power.

Verification. Four rounds by Codex/GPT-6 and Gemini (documenti/verifiche/), with each of their numbers rechecked by code:

- **Round 1:** the numbers of the test were correct; the rules of the prediction had to be closed.
- **Round 2:** found a real error: the confirmation could not be reached in the program.
- **Round 3:** both reviewers found the decision core correct.
- **Round 4:** Gemini «correct»; Codex found the core correct plus six robustness defects. These were closed in version 5, and 71 edge-case tests pass, including Codex's exact counterexamples.

Registrations and timestamps.

- **Before the calculation:** the procedure, the data and the program of the test.
- **After it:** outcome notes, logs and revised versions of the prediction, each with its own time.
- **Before any target data:** the rules of the prediction.
- **Public timestamp:** the SHA-256 fingerprints and UTC times (documenti/registri/) were recorded locally, not publicly. The public timestamp will be the publication date of this record; whether it precedes the target data has to be checked at that time.

6. Files

- `codice/`: test_mm_verlinde.py (the registered test), decidi_previsione_v5.py (the decision program), prove_decidi_v5.py (71 edge cases), ricontrollo_montecarlo.py, griglia_estesa.py, ricontrollo_numeri_astra.py (rechecks), test_catene.py (reads the published chains, needs pandas; the archives are listed in SOURCES.md), ricalcolo_catene.py (its last step from the stored moments), ricalcolo_censimento_v2.py (the census recomputed), prova_corsa3_buttazzo_v2.py (the running against Buttazzo et al.), run_all.py.
- `dati/`: PROVA_FORMATO_DATI_2025_v4.json, the 2025 data in the program's input format (a format check, not a test); covarianze_catene.json, the weighted means and covariances of ω_b, ω_c, h from the chains.
- `risultati/`: reference outputs of all programs.
- `documenti/censimento/`: the Italian working catalogue v3.
- `documenti/test_oggi/`: procedure, data and deviation, outcome notes; `catene/`: the chain check (plan, file list, fingerprints, log, outcome note, verification; in Italian).
- `documenti/previsione/`: the five versions of the prediction.

- `documenti/verifiche/`: the reports of the four rounds.
- `documenti/ineditezza/`: the arXiv API responses of 16.10.2026, 01:49 UTC.
- `documenti/forte/`: the dependency graph of the force-sector readings (which pass through R), the declaration that α_s is imported, and the sensitivity check of $\alpha_3 = (\alpha_2 + 2\alpha_Y)/3$ to α_s through the running (in Italian).
- `documenti/forte/test_G2/`: the registered prospective test of $\alpha_3 = (\alpha_2 + 2\alpha_Y)/3$ on the next PDG edition and the next FLAG review (versions 1-3, verification reports, inputs; in Italian). Program: codice/test_g2_v3.py with the Standard Model running codice/corsa3.py (Buttazzo et al. 2013); 27 edge cases in codice/prove_g2_v3.py.
- `documenti/verifiche_publish/`: the four checks before publication (outcome note in Italian, logs, registrations, fingerprints of the INSPIRE, arXiv and Zenodo answers).
- `documenti/registri/`: the registration lines (UTC time and SHA-256) of the registered files.
- `SOURCES.md`: the measured values used, with their exact tables and equations.
- `IMPRONTE.sha256`: fingerprints of all files.

How to run. `python codice/run_all.py` reruns every program and compares its output with the reference files. It needs Python 3.9 or later, numpy and scipy, and takes under a minute; before deposit, a run from scratch on the extracted archive matched all ten outputs (the record of that run is kept with the upload files, outside the archive); anyone can repeat it with the command above.

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Works cited in the record, with their DOI.

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